g0-sense — Design Document

ai-ee v1 pipeline

generated 2026-08-27 13:20:40

Contents

1 Board and Run Metadata

Field	Value
board	g0-sense
workspace	`boards/g0-sense`
phase	P4
gate status	all 1 recorded gates pass
state created	2026-08-27T06:35:38
state updated	2026-08-27T13:18:16

2 Overview

Brief: g0-sense (unattended container run, 2026-08-27)

A small 2-layer USB-C powered temperature/humidity sensor node - a product-scope board (no mode token: design normally), meant to be sent to JLCPCB as-is.

- Power: USB-C receptacle used for POWER ONLY (5 V VBUS; 5.1 k CC pull-downs so any USB-C source or a legacy A-to-C cable enables VBUS; D+/D- unconnected). Sensible input protection for a USB-powered device (VBUS TVS, resettable fuse or equivalent, reverse/inrush as the architect sees fit).
- 3.3 V rail from a fixed LDO, >= 300 mA, JLC Basic/Preferred part if one fits (AMS1117-3.3 class is fine), with proper input/output capacitors.
- MCU: STM32G030F6P6 (TSSOP-20) on the internal oscillator - no crystal. Decoupling per the datasheet, NRST with a push button, BOOT0 handled.

- Sensor: Sensirion SHT40 (or the SHT4x family member in JLC stock) on I2C with pull-ups; give it a thermal isolation slot/cutout or at least keep it away from the LDO and MCU heat.
- Interfaces: 4-pin SWD header (0.1 in), 4-pin UART header (0.1 in: GND, 3V3, TX, RX) for readout/logging, one Qwiic/STEMMA-QT connector (JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL) sharing the sensor I2C bus.
- One user LED (GPIO, series resistor) and one power LED.
- 2 layers, 1.6 mm FR-4, HASL, green; JLC economy PCBA for the SMD parts, prototype qty 5. Prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).
- Size: whatever an honest layout wants, roughly 35 x 25 mm or smaller; four M2 mounting holes if they fit without hurting the layout.
- No battery, no mains, no USB data, no RF.

Stackup

Chosen stackup: `JLC2313_1.6` (reference/stackups.yaml, `available: true`)

3 Requirements

requirements - g0-sense

P0 requirements distilled from `brief/brief.md` (2026-08-27, unattended container run).

1. Function

Mode declaration: the brief carries NO mode token, so mode = none - design normally. Scope = product-scope per the brief's own words ("a product-scope board ... meant to be sent to JLCPCB as-is"): protection, filtering, connectors, thermal and enclosure-fit are all in scope, and their absence is a reviewer finding. Binding: the stated ~35 x 25 mm size is a SOFT preference, not a hard cap (see section 5). Stage under study: none.

The board is a small 2-layer USB-C powered temperature/humidity sensor node. A USB-C receptacle supplies 5 V (power only, no data); a fixed 3.3 V LDO powers an STM32G030F6P6 MCU running on its internal oscillator, which reads a Sensirion SHT4x sensor over I2C. The same I2C bus is exposed on a Qwiic/STEMMA-QT connector; readout/logging is over a 4-pin UART header; programming/debug is over a 4-pin SWD header. One user LED and one power LED. Prototype quantity 5, JLC economy PCBA, meant to be ordered as-is.

2. Interfaces

- USB-C receptacle, POWER ONLY: 5 V VBUS input; 5.1 k pull-downs on CC1 and CC2 (so any USB-C source or a legacy A-to-C cable enables VBUS); D+/D- unconnected; no USB data.
- SWD header: 4-pin, 0.1 in pitch. ASSUMED: pins GND, 3V3, SWDIO, SWCLK (exact order per architect; brief states only "4-pin SWD header (0.1 in)").
- UART header: 4-pin, 0.1 in pitch: GND, 3V3, TX, RX - for readout/logging. ASSUMED: TX/RX are named from the MCU's perspective (TX = MCU transmit).
- Qwiic/STEMMA-QT connector: JST SH 1.0 mm 4-pin, GND/3V3/SDA/SCL, sharing the sensor I2C bus. I2C pull-ups required on the board (values per architect).

- User LED: one, driven from a GPIO through a series resistor.
- Power LED: one (which rail it indicates - VBUS or 3V3 - is per architect; the brief does not say).
- NRST push button (momentary reset).
- BOOT0: handled (method - strap, option bits, or jumper - per architect; the brief requires only that it is handled).
- Four M2 mounting holes, CONDITIONAL: only if they fit without hurting the layout (see section 5).

3. Power

- Input: USB-C VBUS, 5 V nominal, device draws < 1 A total. No battery, no charging, no mains.
- VBUS input protection required (product scope, brief-stated): TVS, resettable fuse or equivalent; reverse/inrush handling as the architect sees fit - the brief delegates the exact topology.
- 3.3 V rail: fixed LDO, >= 300 mA. JLC Basic/Preferred part preferred; AMS1117-3.3 class acceptable. Proper input/output capacitors per the chosen part's datasheet.
- Rail budget (GUESSES - to be replaced by the architect's numbers):
  - STM32G030F6P6 on internal oscillator: ~5-15 mA (guess)
  - SHT4x: < 1 mA average, single-digit mA peaks during measurement (guess)
  - LEDs: ~2-5 mA total (guess)
  - Qwiic downstream devices: not budgeted by the brief - see open question 2 (recommended default: reserve 100 mA).
  - On-board total is tens of mA; the brief's >= 300 mA LDO floor leaves the headroom for Qwiic loads.

4. Environment

Not stated in the brief. ASSUMED: indoor, room-ambient use (roughly 0-40 C), bare board, no ingress or vibration requirements (see open question 1). The one stated thermal/environmental requirement is layout-level: the SHT4x gets a thermal isolation slot/cutout, or at minimum is kept away from LDO and MCU heat, so self-heating does not corrupt the measurement.

5. Size & mounting

- Outline: "whatever an honest layout wants, roughly 35 x 25 mm or smaller". This is a SOFT preference - no HARD cap. It must NOT bind at P5 board_init; the layout earns the final outline, and ~35 x 25 mm is a target to aim for, not a box to squeeze into (see open question 5).
- Mounting: four M2 mounting holes, CONDITIONAL - "if they fit without hurting the layout". The brief itself allows dropping or reducing them in favor of layout honesty; record that decision if taken.
- Height: no limit stated.

6. Quantity & budget

- Quantity: 5 boards, prototype run, assembled.
- Unit cost: no target stated (see open question 3). Cost posture from the brief: JLC economy PCBA; prefer JLC Basic parts; Extended parts only where the brief names the part (MCU, sensor, connectors).

7. Assembly

- JLC economy PCBA for the SMD parts.
- ASSUMED: single-sided SMD assembly (economy PCBA norm, and this part count does not need the second side).
- Through-hole parts: economy PCBA assembles SMD only, so the two 0.1 in headers are ASSUMED to ship unpopulated for owner hand-soldering (see open question 4).
- ASSUMED: NRST push button and the USB-C / JST SH connectors are SMD, JLC-assemblable variants where stock allows, so they ride the PCBA run.
- Board: 2 layers, 1.6 mm FR-4, HASL, green (brief-stated).

8. Compliance/safety flags

Each flag checked against the brief; none applies:

- Mains voltage: NOT APPLICABLE - brief states "no mains"; input is USB 5 V.
- Battery: NOT APPLICABLE - brief states "no battery"; no charging circuitry anywhere in scope.
- Motors: NOT APPLICABLE - no motor or actuator in the brief's scope.
- >30 V: NOT APPLICABLE - the highest voltage on the board is USB VBUS at 5 V nominal.
- High current (>3 A): NOT APPLICABLE - the brief describes a USB 5 V, < 1 A device.
- RF transmit: NOT APPLICABLE - brief states "no RF"; no radio, and no USB data either ("D+/D- unconnected").

9. Open questions

Closed-form, each with a recommended default. Unattended run: the orchestrator answers these on the owner's behalf and records the answers.

1. Operating environment: is indoor, room-ambient use (about 0-40 C, bare board, no ingress/vibration requirements) correct? Recommended default: YES. 2. Qwiic 3.3 V budget: how much current may external Qwiic/STEMMA-QT devices draw from the 3.3 V rail? Recommended default: reserve 100 mA (fits under the >= 300 mA LDO floor together with the on-board load). 3. Unit cost: is there a hard per-board cost cap? Recommended default: NO hard cap - minimize via economy PCBA and Basic parts as the brief already directs. 4. Header population: should the two 0.1 in headers (SWD, UART) ship unpopulated for hand-soldering, given economy PCBA assembles SMD only? Recommended default: YES - footprints and holes present, headers not populated at assembly. 5. Size confirmation: is ~35 x 25 mm truly a soft preference (the layout may exceed it if honesty demands), not an enclosure-driven hard cap? Recommended default: SOFT preference, per the brief's own wording ("whatever an honest layout wants").

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	unattended default: environment = indoor bench/room ambient, 0-40 C operating, bare board, no ingress or vibration rating	Q1 of requirements section 9; brief names no environment. Most conservative common-practice choice for a USB-powered indoor sensor node; parts will still be picked at industrial -40..85 C grade where that is the stocked option, so the assumption cannot make the board wrong.
P0	unattended default: Qwiic/STEMMA-QT downstream budget = 100 mA reserved on the 3V3 rail	Q2; brief does not budget it. 100 mA covers typical Qwiic sensor/display loads and still fits inside the brief's >= 300 mA LDO floor with margin for the on-board tens of mA, so it sizes the LDO and its thermals honestly without inflating the part.
P0	unattended default: no hard per-board cost cap; minimize cost via JLC economy PCBA and Basic parts as the brief already directs	Q3; brief states no budget. Cost is already bounded by the brief's own Basic-parts-preferred rule; inventing a cap could force a worse engineering choice.
P0	unattended default: the two 0.1 in headers (SWD, UART) ship UNPOPULATED - plated through-holes only, hand-solderable; only SMD parts go to JLC economy PCBA	Q4; brief says 'JLC economy PCBA for the SMD parts'. JLC economy assembly is SMT-only, so THT headers are BOM 'do not populate / customer supplied'. Standard practice for dev/sensor boards and it keeps the assembly quote in economy tier.
P0	unattended default: ~35 x 25 mm is a SOFT preference, not an enclosure-driven hard cap; no HARD outline cap binds at P5	Q5; the brief says 'whatever an honest layout wants, roughly 35 x 25 mm or smaller' and names no enclosure. Confirms requirements.md section 5; placement must not optimize to fit an unearned number (the bb-buck failure mode).
P1	unattended default: LDO = AMS1117-3.3 class in SOT-223 (JLC Basic, C6186); governing thermal design point is the brief's literal >=300 mA rated case (0.51 W), not the ~150 mA realistic load	research/ldo.md escalated the SOT-223-vs-SOT-23-5 tradeoff; research/power.md resolved it with numbers: AMS1117 Tj 71-86 C at 40 C ambient and <=115 C even in the 500 mA abuse case, while AP2112K-class SOT-23-5 (theta-JA 184 C/W) hits Tj 134 C at the rated case with no copper fix available. The Qwiic port is an unfenced 3V3 output, so the rated case is reachable in the field. Basic part, 1.1M stock, $0.20.
P1	unattended default: a sub-6 V TVS clamping voltage is NOT a requirement; SMAJ5.0A/SMF5.0A-class (standoff 5.0-5.5 V, Vc ~9.2 V at Ipp) is accepted	research/protection.md flagged that no passive TVS clamps under 6 V. Nothing on VBUS is 6 V-rated: the only VBUS consumers are the LDO (AMS1117 Vin abs max 15 V), the input caps (16 V+ parts) and the PTC (16 V). Everything at 3.3 V sits behind the LDO. A ~9.2 V clamp during a surge is therefore inside every VBUS part's rating - the original <6 V framing was an over-tight derived spec, not a real one.

Phase	What	Why
P1	unattended default: NO series reverse-polarity element (no Schottky, no P-FET ideal diode) on VBUS	Two independent reasons, both recorded in research: (a) a compliant USB-C cable cannot present reverse polarity (protection.md), and (b) research/power.md computed the end-to-end dropout budget and found a series Schottky's 125-190 mV would consume the entire AMS1117 headroom at the 4.75 V source corner (+0.11 V worst-corner margin as designed). The unidirectional TVS already crowbars a reverse fault. Adding it would make the board worse, not safer.
P1	unattended default: USB-C receptacle = 16-pin power-only part whose THT shield/mounting legs are accepted; JLC assembles it in the Economic PCBA tier	research/connectors.md correctly flagged that every 16P part in this class has THT shield legs and asked whether economy PCBA can place them. Verified directly on the JLCPCB part page for C165948 (TYPE-C-31-M-12): Assembly Process = SMT, Mount Type = Surface Mount Right Angle, and the part is listed as supported by BOTH Economic and Standard assembly. So the legs are not an assembly blocker. Part is JLC Extended - permitted, the brief names the connector.
P1	unattended default: SHT4x on-chip heater (75 mA peak) IS budgeted in the rail tree at <=10% duty	research/power.md's call, accepted: including it costs nothing (peak +3V3 still 185 mA, 240 mA with 30% headroom, under the 300 mA LDO floor and at 39% of the 500 mA CC-blind USB entitlement), while omitting it would turn a documented firmware feature into a hardware ECO later.
P1	unattended default: NRST tactile button - prefer the JLC Basic 5.1x5.1 mm part over the brief's smaller 3x2.5/4.2x3.2 mm size class if the layout has room; size class is a preference, Basic status and stock depth are the tiebreak	research/connectors.md Q2: the only Basic button is 5.1x5.1 mm (C318884, 962k stock, $0.02); the closest-size alternative carries a documented lib_pull symbol gotcha in this repo's own LEARNINGS. The brief never fixed the button size, and the board size is soft, so paying ~2.9 mm of edge for a Basic, gotcha-free part is the conservative trade. Architect may overrule at P2 if placement suffers.
P1	unattended default: LED colours - user LED green, power LED red; both 0603, series resistors sized from the measured Vf band per colour	research/connectors.md Q3 left the assignment open. Red is the conventional power indicator and is the JLC Basic option with the low Vf band (1.8-2.4 V), which gives the largest resistor and the lowest current draw for an always-on LED. Green for the user LED is the conventional 'activity' colour; it is Extended in the KT-0603 family, which is acceptable at one extra unique part, and its 2.6-3.1 V Vf is fine from 3.3 V at a few mA.
P2	research task block-mcu-1 closed over 1 unverified draft record(s): mcu-boot-strap-only-when-option-bytes-consult-pin	–accept-drafts: the slot ships without this knowledge

Phase	What	Why
P2	unattended default: BOOT0 gets a populated 10k pull-down on PA14 (pin 19, shared with SWCLK) - the board does NOT rely on the unverified factory-option-byte claim	The research record mcu-boot-strap-only-when-option-bytes-consult-pin was REFUTED by a fresh second reader: its load-bearing claim (factory 0xDFFFE1AA / nBOOT_SEL=1 makes the BOOT0 pin ignored) rests solely on a community.st.com forum page, and RM0454/AN2606 could not be acquired - st.com times out from this container (verified twice by curl, once by an agent). Rather than design on unverified knowledge, the board is made correct under EITHER option-byte state: a 10k pull-down guarantees BOOT0=0 at reset so the part boots main flash even if nBOOT_SEL=0. Cost is one 0402 resistor; 330 uA load on SWCLK is negligible to any debugger's push-pull driver, and the datasheet-confirmed internal pull-down on PA14 in debug mode already pulls the same way. This closes the gap without the source.
P2	unattended default: I2C pull-ups are 1.5 kOhm (not the 2.2 kOhm the P1 fragment proposed)	The draft record sht4x-i2c-pullup-bus-cap was refuted for recommending 2.2k over its own declared 150-200 pF bus budget: UM10204 Eq 1 at tr=300 ns gives Rp(max) = 354/Cb[pF] kOhm, i.e. 1.77k at 200 pF, so 2.2k needs Cb <= 161 pF and misses the Fast-mode rise time on a cabled Qwiic bus. 1.5k holds the ceiling to 236 pF and clears both floors (967 Ohm from UM10204 Table 10 at 3.3 V/3 mA, 390 Ohm from the SHT4x datasheet). Value independently recomputed by a second reader. architecture/constraints.json and the P4 schematic must use 1.5k.
P3	unattended default: AMS1117 SOT-223 tab is treated as VOUT (+3V3) - the thermal pour is a +3V3 pour, not an isolated island	The datasheet-extractor honestly refused to assert it: the 'TAB IS OUTPUT' label on page 1 sits against the TO-252 view, and no text restates it for SOT-223. Resolved as tab=VOUT because (a) the datasheet's own 15 C/W junction-to-TAB figure means the tab is the die-attach pad, which on this family's leadframe is the VOUT lead, (b) the KiCad standard symbol/footprint pair the librarian pulls ties tab to VOUT, so schematic and layout agree by construction, and (c) it is the safe direction either way: if the tab were floating, a +3V3 pour on it is still harmless, whereas assuming 'floating' and isolating the copper would throw away the thermal design the 0.51 W design point depends on. Flagged for the schematic reviewer to confirm against the pulled symbol.

Phase	What	Why
P3	unattended default: the SHT4x 20 V/ms VDD slew limit needs NO extra bulk capacitance - carried item CLOSED, not escalated	Two independent second readers read the SHT4x datasheet Table 4 slew row and quoted its condition verbatim: 'Voltage changes on the supply between VDD,min and VDD,max. Faster slew rates may lead to a reset' (min cell empty). That bounds supply changes DURING operation, not the cold-start ramp from 0 V - and a reset during power-on is the intended behaviour anyway (the part has a POR). In operation the 22 uF tantalum on +3V3 holds any load-step slew orders of magnitude below 20 V/ms. Adding bulk to slow a cold start would also fight the USB-C <=10 uF attach limit upstream. No hardware change; note it for bring-up.
P4	unattended default: WAIVE schematic-review W1 (no ESD array on the J2 Qwiic SDA/SCL) - ship the board without a TVS/ESD array on the I2C bus	Product-scope connector checklist allows met-or-waived. Waived on four grounds: (a) the Qwiic/STEMMA-QT ecosystem norm (SparkFun/Adafruit reference boards) is a bare 4-pin JST SH with no array, so a user-facing part behaves as the ecosystem's users expect; (b) STM32G030 PA11/PA12 and the SHT40 both carry internal ESD structures and are 2 kV HBM rated, which is the standard handling rating for an indoor bench device; (c) the recorded environment is indoor bench/room ambient with no ingress rating, not a field-exposed port; (d) adding a 4-channel array now means a new Extended MPN plus setup fee, a fresh library pull and footprint verify, and re-opening a green schematic at P4 - real regression risk against a warning-severity, field-reliability-only exposure that cannot prevent bring-up. Recorded in reports/erc-waivers.md and shown at H2.

Phase	What	Why
P4	unattended default: CLOSE the carried verify item 'SHT4x VDD slew <= 20 V/ms vs AMS1117 startup ramp' (schematic-review W2) as accepted risk - no hardware change	Three findings close it. (1) PROVENANCE IS ON FILE, contrary to the reviewer's note: the 20 V/ms figure is research/records/sht4x-vdd-slew-power-up.yaml, sourced to research/sources/HT_DS_Datasheet_SHT4x.pdf p9 Table 4 and read visually TWICE (researcher + fresh second reader, maturity verified). The reviewer grepped parts/C2909890.json (the extraction JSON) and the LCSC PDF, which is a different artifact. (2) THE WORST-CASE NUMBER IS AN UPPER BOUND THAT CANNOT OCCUR: I_limit/C (~1 A into ~27 uF, 40-68 V/ms) assumes the LDO sources its full current limit into the output cap independently of its input. The AMS1117 is a series pass follower - during a cold start VOUT cannot rise faster than VIN minus dropout, so +3V3's edge is set by the USB source's VBUS ramp through C2 = 10 uF, not by the LDO's current limit. (3) THE NAMED FAILURE MODE IS BENIGN AT COLD START: the datasheet says only 'faster slew rates MAY lead to a reset'; the second reader explicitly flagged 'or hung until re-powered' as an unsupported clause the researcher added. A reset during the power-up ramp lands the part in exactly the state a power-up is supposed to produce - there is no state to lose at t=0. The limit's real bite is in-operation supply excursions and hot-plugging a DOWNSTREAM breakout onto a live Qwiic rail, which is the breakout's decoupling problem, not this board's. HARDWARE ALTERNATIVES REJECTED: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak would drop 7.5 V across the 100 ohm needed for a 100 us tau); raising C3 above ~50 uF to force I_limit/C under 20 V/ms buys nothing against (2) and costs a part change on a green schematic. BELT AND BRACES (documentation, not copper): the firmware note 'issue an SHT4x soft reset (0x94) at init, per Sensirion's own recommended start-up sequence' goes into the design doc and fab/README, and the Qwiic hot-plug caveat is recorded as a known limit.

Phase	What	Why
P4	unattended default: carry a P9 REQUIRED check - CPL rotation must be validated for every polarized 2-pin part (D1 TVS SOD-123, D2 power LED, D10 user LED, C3 tantalum CASE-A), not only for J1	schematic-review OPEN item. decisions.md #13 carried a CPL rotation check for J1 alone; the KiCad<->JLC 0-degree-reference offset (reference/jlc_rotations.csv, LEARNINGS/S8) bites every polarized package, and a reversed tantalum on 3V3 or a reversed TVS on VBUS is a dead board, not a cosmetic error. dfm_check.py owns polarity/rotation validation - at P9 its findings for D1/D2/D10/C3 must be read explicitly and each confirmed against the footprint pin-1 marking, not just accepted as a count.

Sheet plan

sheets - g0-sense hierarchical plan (P2)

Small board: root + TWO child sheets. One sheet would also fit, but the power/main split gives P6 its natural placement groups (power entry corridor vs logic/sensor), keeps the Type-C + protection cluster reviewable in one frame, and costs nothing. Refdes are unique ACROSS sheets by disjoint ranges; every range includes a #PWR base.

Root: `g0-sense.kicad_sch`

Holds the two sheet symbols only. Global power nets (power symbols, bare netlist names): `VBUS`, `+5V`, `+3V3`, `GND`. Power symbols make these names bare in the final netlist; all constraints.json net references use the bare forms.

Sheet 1: `power.kicad_sch` (blocks B1 usb-input+protection, B2 ldo-3v3)

- Contents: J1 USB-C receptacle; R1, R2 (5.1k Rd); D1 (TVS); F1 (PTC); C1 (100 nF VBUS); U1 (AMS1117-3.3); C2 (10 uF X5R, +5V at VIN); C3 (22 uF tantalum, +3V3); D2 (power LED red) + R3 (620 Ohm).
- Interface nets (hier pins -> placement groups at P6): `+3V3` (out), `GND`. VBUS and +5V stay sheet-local (they leave via power symbols, so their netlist names are still bare `VBUS` / `+5V`).
- Sheet-local labels: `CC1`, `CC2` (final names `/power/CC1`, `/power/CC2`).
- Refdes ranges: J1; U1; F1; D1-D9; R1-R9; C1-C9; SW none; #PWR pwr_base=100 (#PWR0100-#PWR0199).

Sheet 2: `main.kicad_sch` (blocks B3 mcu, B4 sht4x + I2C/Qwiic, headers)

- Contents: U2 (STM32G030F6P6); U3 (SHT40-AD1B class); C10 (100 nF U2), C11 (4.7 uF U2), C12 (100 nF NRST), C13 (100 nF U3); R10, R11 (1.5k I2C pull-ups), R12 (220 Ohm user LED), R13 (10k BOOT0 pull-down); D10 (user LED green); SW1 (NRST button); J2 (Qwiic JST SH); J3 (SWD 1x4 DNP); J4 (UART 1x4 DNP); H1-H4 (M2 mounting holes, conditional).
- Interface nets: `+3V3` (in), `GND`.
- Sheet-local labels (final names `/main/<label>`): `SDA`, `SCL`, `SWDIO`, `SWCLK`, `UART_TX`, `UART_RX`, `NRST`, `LED_USER`.
- Refdes ranges: U2-U9; J2-J9; D10-D19; R10-R19; C10-C19; SW1-SW9; H1-H9; #PWR pwr_base=200 (#PWR0200-#PWR0299).

Interface-net contract (P4 must produce exactly these)

Net (netlist form)	Class	From	To
VBUS	power (bare)	J1	D1, F1, C1
+5V	power (bare)	F1	U1 VIN, C2
+3V3	power (bare)	U1 VOUT	C3, D2/R3, U2, U3, R10, R11, J2, J3.2, J4.2
GND	power (bare)	all	all + J1 shield
/power/CC1, /power/CC2	signal	J1 CC pins	R1, R2
/main/SDA, /main/SCL	signal	U2 PA10/PA9	U3, J2, R10/R11
/main/SWDIO, /main/SWCLK	signal	U2 PA13/PA14	J3.3/J3.4; SWCLK also R13 -> GND
/main/UART_TX, /main/UART_RX	signal	U2 PA2/PA3	J4.3/J4.4
/main/NRST	signal	U2 pin 6	C12, SW1
/main/LED_USER	signal	U2 PA5	R12 -> D10

Header pinouts (canonical): J3 SWD = 1:GND, 2:3V3, 3:SWDIO, 4:SWCLK. J4 UART = 1:GND, 2:3V3, 3:TX (MCU transmit), 4:RX (MCU receive). Pin 1 marked on silk on both; no vendor standard exists for a 4-pin SWD header, so the silk labels are the contract.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 2 attempt(s); last 2026-08-27T13:18:16, failing 0 of 0.

Review waivers

P4 schematic waivers - g0-sense

ERC gate itself is 0 errors / 0 warnings (reports/erc.json), and `netlist_audit` (log/netlist_audit-P4.json) is 0 violations with 99/99 expected pins connected. Nothing below is an ERC violation: these are the three WARNINGS raised by the fresh-context `schematic-reviewer` (reports/review-schematic.{md,json}), with the disposition taken on the owner's behalf under the unattended run contract. 0 errors were raised.

#	kind	refs	severity	disposition
W1	qwiic-esd-unwaived	J2, U2, U3	warning	WAIVED (recorded decision, P4)
W2	sht4x-vdd-slew-unclosed	U3, U1, C3	warning	CLOSED as accepted risk (recorded decision, P4)
W3	dnp-not-native	J3, J4	warning	FIXED (work order wo-w3-dnp)

W1 - no ESD array on the J2 Qwiic SDA/SCL - WAIVED

Finding: J2 is a populated, user-facing, hot-pluggable connector; SDA/SCL and +3V3 leave the board with no clamps and no series impedance, so a cable-borne strike reaches U2 PA11/PA12 and U3 (2 kV HBM) directly. The product-scope connector checklist requires this expectation met OR waived, and no waiver existed. Field-reliability only - it cannot prevent bring-up.

Waived on four grounds:

1. Ecosystem norm. The Qwiic / STEMMA-QT reference implementations (SparkFun, Adafruit) are a bare 4-pin JST SH with no array. A user-facing part that behaves the way the ecosystem's users expect is the conservative choice here, not the exceptional one. 2. Both silicon endpoints carry internal ESD structures and are rated 2 kV HBM - the standard handling rating for an indoor bench device. 3. The recorded operating environment (P0 decision) is indoor bench / room ambient, bare board, no ingress rating - not a field-exposed port. 4. Cost of the alternative at

this point in the run: a new Extended MPN plus its JLC setup fee, a fresh library pull + fp_verify, and re-opening a green schematic - real regression risk traded against a warning-severity exposure.

If the owner wants the array anyway, the cheap retrofit is a 4-channel low-capacitance array on SDA/SCL at J2 (the 1.5 k pull-up budget has room: the bus was sized to 236 pF and an array adds single-digit pF). That is a post-run `add-part` task, not a rewind.

W2 - carried SHT4x VDD-slew verify item - CLOSED as accepted risk

Finding: `power_tree.md` / `decisions.md` carried "verify SHT4x VDD slew <= 20 V/ms vs the AMS1117 startup ramp"; the reviewer found it unclosed, found no slew spec in `parts/C2909890.json`, and computed ~40 V/ms from AMS1117 current-limit charging of the ~27 uF on +3V3.

Closed, no hardware change, on three findings:

1. **Provenance is on file** (the reviewer looked in the wrong artifact). The number is `research/records/sht4x-vdd-slew-power-up.yaml`, sourced to `research/sources/HT_DS_Datasheet_SHT4x.pdf` p9 Table 4, read visually by the researcher and again by a fresh second reader (`maturity: verified`). `parts/C2909890.json` is the P3 extraction, a different artifact with a different scope. 2. **The worst-case number is an upper bound that cannot occur.** I/C (~1 A into ~27 uF) assumes the LDO drives its output cap independently of its input. The AMS1117 is a series-pass follower: on a cold start VOUT cannot rise faster than VIN minus dropout, so the +3V3 edge is set by the USB source's VBUS ramp filtered by C2 = 10 uF, not by the LDO's current limit. 3. **The named failure mode is benign at cold start.** The datasheet says only "faster slew rates MAY lead to a reset" - and the second reader already flagged the researcher's added "or hung until re-powered" as unsupported by any cited page. A reset during the power-up ramp leaves the part in exactly the state a power-up is meant to produce. The limit's real bite is in-operation supply excursions, and hot-plugging a DOWNSTREAM breakout onto an already-live Qwiic rail - that one is the breakout's decoupling problem, not this board's.

Hardware alternatives considered and rejected: a series R at U3 VDD to slow the local ramp is disqualified by the SHT4x on-chip heater (75 mA peak across the ~100 ohm needed for a 100 us tau = 7.5 V drop); raising C3 past ~50 uF to force I/C under 20 V/ms buys nothing against finding 2 and costs a part change on a green schematic.

Belt and braces, documentation only: "issue an SHT4x soft reset (0x94) at init" (Sensirion's own recommended start-up sequence) goes into the design doc and `fab/README`, and the Qwiic hot-plug caveat is carried as a known limit of the board.

W3 - J3/J4 DNP intent not KiCad-native - FIXED

Not waived. The headers carried DNP only in a custom `Variant=DNP` field and the value string, so a native KiCad BOM/POS export would have emitted the two THT headers as populate lines into a JLC economy (SMT-only) order. Dispatched as work order `log/workorders/wo-w3-dnp.json`: native `dnp` attribute set through the board's own generator, and `refdes_class: {J3, J4 -> hand_install}` added to the header line in `parts/parts.json` so `bom_cpl.py` keeps them out of the assembler upload and the CPL while still listing them in the documentation BOM. ERC re-run after the change.

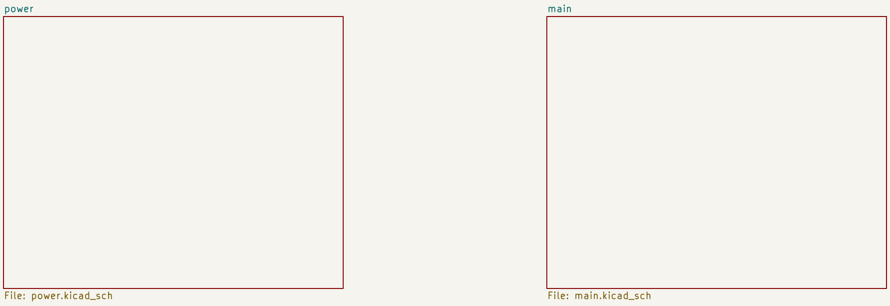
Reviewer OPEN items and where they went

- **AMS1117 SOT-223 tab = VOUT not stated in the on-file PDF.** Already a recorded P3 decision (the pulled symbol's pin 4 = VOUT, and SOT-223 construction fuses the tab to the pin-2 lead frame in every 1117-family part). No further action.
- **SW1 pairing verified from a live fetch, not archived.** The evidence (A/B common, C/D common; NRST on 1/2, GND on 3/4) is recorded in `reports/review-schematic.md`

with its source URL and is committed. Not re-fetched: the finding is a CLEAR, and the archive would only make an already-recorded pass reproducible.

- **Extend the CPL rotation check from J1 to the polarized 2-pin parts (D1, D2, D10) and to C3.** Carried into P9 as a required check; recorded as a decision so it cannot be lost.

The full schematic PDF follows.



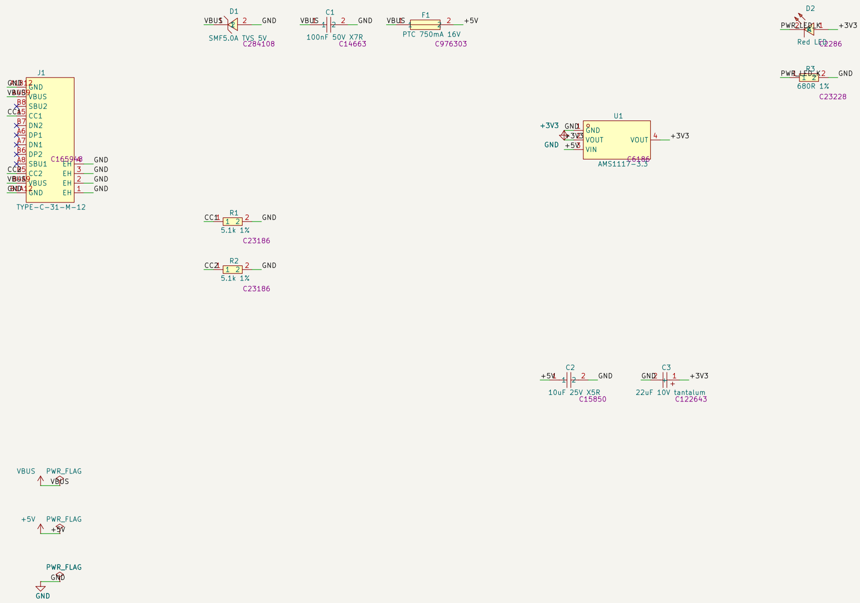
ai-ee

Sheet: /

File: g0-sense.kicad_sch

Title: g0-sense: USB-C powered STM32G0 + SHT40 sensor node

Size: A4	Date: 2026-08-27	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 1/3



ai-ee
Sheet: /power/
File: power.kicad_sch

Title: g0-sense: USB-C power entry + 3V3 LDO

Size: A3 | Date: 2026-08-27 | Rev: 1
KiCad E.D.A. 10.0.5 | Id: 2/3

R10 1.5k C22843 SDA +3V3
R11 1.5k C22843 SCL +3V3
C12 100nF 50V X7R C14663 NRST GND
Tactile button
SW1 C318884 NRST GND
R13 10k C25804 SWCLK GND
R12 100R C22775 LED_USER LED_A
D10 Green 0805 C2297 GND LED_A
C10 100nF 50V X7R C14663 +3V3 GND
C11 4.7uF 16V X5R C19666 +3V3 GND
+3V3
GND
U2 STM32G030F6P6 C724040
PB7/PB8 PB9/PC14-OSC32_IN PC15-OSC32_OUT VDD/VDDA VSS/VSSA NRST PA0 PA1 PA2 PA3
PB3/PB4/PB5/PB6 PA15/PA14-BOOT0 PA13 PA12[PA10] PA11[PA9] PB0/PB1/PB2/PA8 PA7 PA6 PA5 PA4
SWCLK SWDIO SDA SCL LED_USER
J3 GND +3V3 SWDIO SWCLK C2713270 1x4 2.54mm THT (DNP)
J4 GND +3V3 UART_TX UART_RX C2713270 1x4 2.54mm THT (DNP)
Qwiic JST SH 4P
J2 C160404 GND +3V3 SDA SCL
U3 SHT40-AD1B-R2 C2909890 SDA SCL EP VSS VDD GND +3V3
C13 100nF 50V X7R C14663 +3V3 GND
ai-ee
Sheet: /main/
File: main.kicad_sch
Title: g0-sense: MCU + SHT40 + I2C/Qwiic + headers
Size: A3 Date: 2026-08-27 Rev: 1
KiCad E.D.A. 10.0.5 Id: 3/3

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - g0-sense (2026-08-27)

- Artifact: ‘requirements.md‘ (9 sections), ‘check_requirements.py‘ PASS (0
violations).
- Mode: NO mode token in the brief -> mode = none, design normally. Scope is
product per the brief’s own words, so absent protection/filtering/connectors/
thermal IS a reviewer finding downstream.
- Geometry: ~35 x 25 mm recorded SOFT (no HARD cap) - must not bind at P5
board_init; four M2 holes conditional on the layout.
- Safety (section 8): mains / battery / motors / >30 V / >3 A / RF transmit
all
NOT APPLICABLE with brief evidence (USB-C 5 V power-only, <1 A). No safety
unknown to escalate; nothing PROVISIONAL on safety grounds.
- 5 OPEN questions answered as unattended defaults and recorded as decisions:
environment indoor 0-40 C; Qwiic budget 100 mA; no hard cost cap; 0.1 in
headers ship unpopulated (economy PCBA is SMT-only); size stays soft.
- Spawn: requirements-analyst @ fable/high (session effort xhigh).
- Next: P1 research roster (power architect, component scout, interface spec).
```

P1

```
# P1 Research digest - g0-sense (2026-08-27)

Roster (8 agents): 4x research-component-scout (ldo, protection, connectors,
mcu-sensor), research-interface-spec (usbc-power-sink), 2x
research-reference-design (sht4x, stm32g030-minimal), research-power-architect.

Load-bearing findings
- USB-C CC-blind sink (fixed 5.1k Rd on CC1 AND CC2, independent - never one
shared resistor) is entitled to DEFAULT USB power only: budget <= 500 mA.
Sink capacitance at the receptacle <= 10 uF (TC2.0 Table 4-3).
- Rail tree: VBUS -> TVS -> PTC -> +5V -> LDO -> +3V3. +3V3 peak 185 mA
(incl. 100 mA Qwiic reserve + 75 mA SHT4x heater), 240 mA with 30% headroom.
VBUS worst draw 196 mA = 39% of entitlement.
- LDO: AMS1117-3.3 SOT-223 (C6186, Basic) beats SOT-23-5 low-Iq parts on the
only axis that decides: Tj 71-86 C vs 134 C at 40 C ambient, 0.51 W case.
Needs a 22 uF TANTALUM-class output cap (not plain ceramic) - carry to P3/P4.
Dropout closes at the 4.75 V corner with +0.11 V worst-case margin.
- STM32G030F6P6 TSSOP-20: VDD/VDDA bonded to pin 4, VSS/VSSA pin 5, no
separate
```

VREF+. BOOT0 needs NO strap - factory nBOOT_SEL=1/nBOOT0=1 always boots main flash, and BOOT0 shares pin 19 with SWCLK anyway. NRST needs only 100 nF + the button (internal pull-up 40k typ). Pin map: SWD PA13/PA14, USART2 PA2/PA3, I2C1 PA9/PA10 (5 V tolerant, Fm+), LED PA5. 9 of 20 pins used, no conflicts.
- SHT4x: 100 nF at VDD, 2.2k I2C pull-ups (sized for a 0.5 m Qwiic leg), copper
conduction is the dominant self-heating path -> distance + necked traces + no pour under the part + a milled slot. NO BOARD WASH (Sensirion forbids it) -> must appear in the fab/assembly notes. Fixed I2C address per variant.
- MCU + sensor are both JLC EXTENDED (no Basic exists in either family): STM32G030F6P6 C724040 $1.37, SHT40-AD1B-R2 C2909890 $1.90 (addr 0x44).

7 OPEN questions answered and recorded as decisions (LDO design point + part; TVS clamp ceiling dropped as an over-tight derived spec; no reverse-polarity element; USB-C THT legs verified OK in JLC Economic tier; heater budgeted; button size vs Basic; LED colours). Gates: none due at P1.

P2

P2 Architecture digest - g0-sense (2026-08-27)

- architect (fable/high) wrote blocks.md / power_tree.md / stackup.md / sheets.md / constraints.json / decisions.md. constraints_lint exit 0.
- 4 blocks: B1 usbc-sink, B2 ldo, B3 mcu, B4 sht4x. Stackup JLC2313_1.6 (2L, 1 oz, HASL). Two sheets (power / main), disjoint refdes + #PWR 100/200.
- No diff pairs, no high-speed nets, no sim bench (recorded, with reasons).
- P2 coverage exit: FIRST run 4 slots / 4 GAP (no checklist existed for any of usbc-sink, ldo, mcu, sht4x). research.py open --all -> 4 tasks (per_run cap 6, 2 left for P3). 4 researchers -> 16 quarantined sources, 4 new checklists, 23 draft records. 4 fresh second readers refuted 6. 5 were repaired against ledger pages and re-read; the NRST record took three repair/re-read cycles (each refutation narrower than the last) before verifying.
- FINAL: 22 of 23 records verified. Re-run coverage: 4 slots, 0 gap, 4 provisional, 1 draft_unverified.
- The one unverifiable record (mcu boot-strap) is a container limit, not a research failure: st.com times out, so RM0454/AN2606 cannot be acquired. Its claim was load-bearing, so the DESIGN was changed to not need it.
- Two research-forced design changes vs the architect's plan: I2C pull-ups 2.2k -> 1.5k (Rp ceiling arithmetic), BOOT0 no-strap -> R13 10k pull-down. architecture/{blocks,sheets,decisions}.md updated to match.
- H1 written to log/H1.md, approved as delegated. report_gen exit 0 (10 pp).

P3

P3 Parts + Library digest - g0-sense (2026-08-27)

- part-sourcer (sonnet/high) -> parts/parts.json: 20 distinct parts covering 26 electrical refdes. 12 Basic / 8 Extended, of which 7 bear a JLC setup fee (the DNP header is Extended-priced but never machine-placed). ~$4.42/board, ~$22.10 for the 5-board run. All stock re-verified live; worst case (MCU C724040, 8351 pcs) is 334x the 5x-build-qty floor.
- Four sourcing corrections applied on review, three of them fee removals: R3 620R Extended -> 680R Basic (620R is a genuine E24 Basic-catalog gap); user LED KT-0603G Extended -> KT-0805G, which IS Basic; R12 220R -> 100R (electrical, not cost: green Vf 2.6-3.1 V against 3.3 V gave 0.9 mA at the worst bin - 100R gives 2-7 mA); whole resistor BOM normalised to 0603.
- 3 datasheet-extractors (parallel) -> parts/C724040.json, C2909890.json, C6186.json, all `datasheet_extract --validate` exit 0. Notable: TSSOP-20 pin 19 is ONE bonded pad carrying PA15 and PA14-BOOT0/SWCLK - transcribed as the datasheet names it rather than picking a function; AMS1117 has NO

```
recommended land pattern (outline only), so that field was omitted rather
than faked; SHT40 die pad is present but explicitly NOT to be soldered.
- librarian (sonnet/medium): 20/20 pulled, 85 pins retyped (ERC prerequisite),
lib tables registered. fp_verify found ONE real defect and it mattered:
the pulled SHT40 DFN-4 soldered the die pad, which would have defeated the
whole thermal-isolation design. Deleted (copper+paste+mask), courtyard
closed. Three more fixes: broken symbol->footprint link (audited all 20 -
the only one); C3 tantalum had NO polarity marking at all - pin 1 = anode
established twice (manufacturer PDF + live EasyEDA CAD data), silk "+" and
bar added; J2 Qwiic pin-1 confirmed against JST's own eSH.pdf ("viewed from
the connector mounting surface", No. 1 circuit leftmost) - no flip needed.
- fp_verify final: 3 passed / 0 failed / 2 accepted pad_size warnings.
- P3 coverage exit: 7 slots, 0 gap, 3 part slots covered, 4 blocks
provisional. No P3 research needed (2 of the 6 per-run tasks still unspent).
- 2 decisions recorded: AMS1117 SOT-223 tab = VOUT (pulled symbol confirms
pin 4 = VOUT); SHT4x 20 V/ms slew item CLOSED, not escalated (the limit is
specified for in-operation supply changes, not the cold-start ramp).
```

P4 *(no digest recorded)*

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-27T09:50:50	erc	pass	0/0
2026-08-27T13:18:16	erc	pass	0/0

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-27T08:44:13	delegated (unattended): architecture + 2L stackup approved; LDO 0.51 W design point approved; boot-strap gap accepted because the 10k BOOT0 pull-down removes the dependency; outline stays an output; no sim bench
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
`state.json`	34,303 B	
`kicad/g0-sense.kicad_sch`	1,710 B	
`brief/brief.md`	1,634 B	
`requirements.md`	6,653 B	
`architecture/blocks.md`	11,799 B	

File	Size	Note
`architecture/constraints.json`	8,406 B	
`architecture/decisions.md`	7,408 B	
`architecture/power_tree.md`	4,040 B	
`architecture/sheets.md`	2,964 B	
`architecture/stackup.md`	2,679 B	
`reports/erc-waivers.md`	6,030 B	
`reports/erc.json`	1,039 B	
`reports/review-schematic.json`	1,621 B	
`reports/review-schematic.md`	7,856 B	
`reports/schematic.pdf`	131,338 B	
`reports/top.net`	60,717 B	